

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct 192511369

I V.C. BHAVANBAIRAV (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.C. Bhavan Bairav

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

A Media company wants to migrate its video streaming platform to the cloud.

cloud Architecture:

- Store Videos in cloud storage.
- Use CDN to cache popular videos.
- Deploy Application Servers with a Load balancer.
- Enable Auto-Scaling for traffic changes.

Scalability & caching:

- Load Balancer distributes user requests evenly.
- Auto-Scaling adds/removes servers automatically.
- CDN reduces buffering by serving cached videos.
- Supports millions of concurrent users.

Cost optimization:

- Use pay-as-you-go pricing.
- Remove idle servers using auto-scaling.
- Store old videos in low cost storage.
- Monitor resource usage regularly.

Recommended Solution:

- Use CDN + Load Balancer + Auto-scaling + cloud storage.
- Ensures high availability, low buffering, better performance, reduced cost.

2) A fintech startup is deploying applications using containers across multiple cloud providers.

1) Container Orchestration & Multi-cloud solution:

- Use Kubernetes to manage containers across multiple cloud providers.
- Deploy the same application on AWS, Azure, Google Cloud.
- Maintain consistent application configuration across all clouds.
- Automate deployment and updates using Kubernetes.

2) Service Discovery & Scaling:

- Use Kubernetes Service Discovery to automatically locate services.
- Enable Auto-Scaling to increase or decrease containers based on workload.
- Use Load Balancing to distribute user requests evenly.

3) Benefits & Risks of Multi-cloud:

Benefits:

- High Availability
- Better Reliability
- Avoids Vendor lock in
- Improved disaster recovery

Risks:

- Complex Management
- Higher Operational cost
- Security and identity management challenges.

4) Solution: Deploy Kubernetes containers across clouds. Use service discovery, auto-scaling and load balancing for efficient management.

An e-learning platform experiences data loss due to accidental deletion in the cloud.

1) Backup and Disaster Recovery Solution:

- Use cloud native backup to automatically back-up data.
- Enable data replication to store copies in different cloud regions.
- Use Versioning to recover deleted or modified Files.
- Schedule regular automatic backups.

2) Achieving RTO and RPO:

- RTO (Recovery Time Objective): Use disaster recovery and failover to restore services quickly.
- RPO (Recovery Point Objective): Use frequent backups and continuous replication to minimize data loss.
- Restore the latest backup when data is deleted.

3) Business Continuity steps:

- Maintain multiple backup copies.
- Replicate data across different regions.
- Regularly test backup and recovery procedures.
- Monitor the backup system continuously.

4) Solution:

- Use cloud-native backup + replication + Versioning.
- This ensures quick data recovery, minimal data loss, and continuous availability of e-learning platform during failures.

4) A government agency is adopting a hybrid cloud model sensitive and non-sensitive data.

1) Hybrid cloud Architecture:

- Store sensitive data in the on-premises data center.
- Store non-sensitive data in the public cloud.
- Connect both environments using a secure VPN or dedicated private connection.
- Allow applications to access data securely between both environments.

2) Identity Management and Encryption:

- Implement Single Sign-on (SSO) for centralized user authentication.
- Use Role-Based Access Control (RBAC) to restrict user access.
- Encrypt data at rest and in transit using secure encryption methods.
- Enable MFA for additional security.

3) Challenges in Hybrid cloud:

- Complex management of 2 environments.
- Security and compliance issues.
- Network latency between on-premises and cloud.
- Difficulty in monitoring and resource management.

4) Recommended Solution:

- Use a hybrid cloud with Secure VPN, SSO, RBAC, MFA and Encryption.

- This ensures secure communication, compliance with government regulations, and efficient management of sensitive & Non-sensitive data.

A SaaS provider needs to ensure high availability and fault tolerance for its application.

1) Multi-Region Deployment Strategy:

Deploy the application in multiple cloud regions.

Use a Load Balancer to distribute traffic between regions.

- If one region fails, traffic is automatically redirected to another region.

- This ensures continuous service availability.

2) Data Replication and Failover:

- Replicate data continuously between regions.

- Maintain a primary and secondary database.

- Configure automatic failover to switch to the backup region during failures.

- This minimizes downtime and data loss.

3) Monitoring and Alerting:

- continuously monitor CPU, memory, network, and application health.

- Configure alerts for server failures, high resource usage, and network issues.

- Use monitoring tools to detect problems early and notify administrators.

4) Recommended Solution:

- Use multi-region deployment, data replication, automatic failover and continuous monitoring.

- This ensures high availability, fault tolerance, quick recovery and uninterrupted SaaS services even if one cloud region goes down.